Summary of Previous Research

Shengxiu Sun | Peking University

My research is organized around one scientific question: how do accreting supermassive black holes and their host galaxies co-evolve? Addressing this question requires measuring two components that are observed together but evolve on very different scales. Nuclear emission can dominate the image and spectrum, while the host records the longer history of star formation, structure, and dynamics. I therefore develop measurements that separate the AGN and host components, quantify the uncertainties of that separation, and use them to make physical inferences. I then apply these tools to several scientific settings, including changing-look AGNs and quasars at cosmic dawn.

Building the measurements needed for AGN–host co-evolution

The first requirement is to recover host-galaxy properties in the presence of a bright nucleus. I developed an iterative spectrophotometric decomposition for 1,083 DESI type-1 quasars at $0.1 < z < 1$ [1]. Subaru/HSC images constrain the host flux within the DESI spectroscopic aperture, while the DESI spectrum constrains the stellar population and AGN emission. The fit iterates between the imaging and spectral information, and the host flux is measured directly from PSF-subtracted images rather than inferred only from an analytic surface-brightness model. This joint approach breaks the AGN–host degeneracy sufficiently to measure host star formation, stellar mass, and black-hole–stellar-mass relations for a large sample.

The scientific result is that the selected extended-quasar hosts are predominantly quiescent, with about 23% showing post-starburst signatures. Thus, substantial black-hole accretion can coexist with weak current star formation. The result also gives a necessary selection statement: extended morphology favours prominent, massive hosts, so this sample is a baseline for comparison with compact quasars rather than a description of the full quasar population. The host stellar populations are consistent with a post-starburst evolutionary history, but the data do not by themselves show that AGN feedback caused the decline in star formation.

This work established both the measurement framework and the host-galaxy baseline for the projects that followed. It also exposed a limitation: host velocity dispersions are difficult to measure when AGN continuum and emission dilute the absorption features. I therefore developed a neural-network method for AGN-contaminated spectra [4]. The network is trained on 220,000 forward-modelled spectra spanning host populations, nuclear contamination, instrumental resolution, wavelength coverage, and observing conditions. It reconstructs the host and AGN spectra while estimating the host stellar velocity dispersion, so the spectral information supporting each estimate can be inspected.

Independent mock tests and comparisons with SDSS and higher-resolution Keck spectra identify the regime in which the method is useful and the regimes in which it is biased. The confidence criteria select spectra with sufficient host information and reconstruction quality, while flagging measurements near the instrumental-resolution limit and at high dispersion. Of 399 observed broad-line AGNs, 159 meet these criteria. Their $M_{\rm BH}$–$\sigma_\star$ slope is consistent with local relations under the adopted uncertainties. I also tested how the result changes with the black-hole mass zero point, error model, and host structure. The result is therefore a practical, uncertainty-aware measurement of a population correlation, rather than an unsupported claim of the intrinsic bulge relation.

Using changing-look AGNs to test the co-evolution picture

A method for separating AGN and host light should be tested in a setting where the two components are known to evolve on different timescales. Changing-look AGNs provide exactly this experiment: the broad lines and continuum can change over years, whereas the stellar population of the host is effectively fixed over the same interval. I extended the decomposition to paired DESI and SDSS spectra for 105 changing-look AGNs [2]. The model shares the underlying stellar population across epochs, allows the

nuclear continuum and broad lines to vary independently, and uses HSC photometry to account for the different fibre apertures.

The two-epoch structure is a scientific constraint, not only a technical detail. A faint state can expose the host, while a bright state constrains the variable nuclear component; consistency between the two states tests whether the decomposition is physically plausible. For 82 extended changing-look AGNs, I constructed a comparison sample of quasars matched in redshift and stellar mass and analysed it with the same pipeline. The post-starburst fractions are 35.4% and 18.3%, respectively, while the quiescent fractions do not differ clearly.

This result addresses both the method and the astrophysics of changing-look AGNs. The higher post-starburst fraction shows that the changing-look population retains information about a distinct host-galaxy history, but the stable stellar component does not support a synchronized, galaxy-wide quenching event. Combined with the rapid nuclear transition, the result favours a change in the central accretion flow as the primary driver of the changing-look phenomenon. CL-AGNs therefore serve two roles in my research: they validate host–AGN decomposition through repeated observations, and they provide a physical laboratory for separating nuclear variability from long-timescale galaxy evolution.

Calibrating the physical scales of the connection

Once the components can be separated, the co-evolution question requires reliable physical scales for both the central black hole and the host. I addressed these scales with complementary near-infrared and optical-spectroscopic projects.

Optical broad lines can be attenuated or unavailable, motivating near-infrared mass estimators. Yet calibrating Paschen lines against another single-epoch estimator transfers that estimator's mass scale into the result. I reduced and analysed LBT/LUCI spectra of 27 SDSS reverberation-mapping quasars and combined them with archival Paschen measurements of objects with reverberation masses [3]. The resulting Pα and Pβ relations are anchored directly to reverberation mapping. Their fiducial intrinsic scatters are 0.28 and 0.32 dex within the calibration sample. I compared line and continuum luminosities, alternative line-profile models, and free or fixed virial exponents; the alternatives overlap within their uncertainties. The contribution is an empirically tested near-infrared black-hole mass scale with explicit measurement definitions and a known domain of application, while its absolute normalization remains subject to the reverberation-mapping virial factor.

The velocity-dispersion work complements this mass calibration by addressing the host side of the relation. The network reconstructs the stellar and AGN components and supplies a confidence assessment rather than a point estimate alone. This makes it possible to identify which spectra genuinely contain the information needed to test black-hole–host scaling relations. Across the two projects, my aim is not simply to obtain a mass or a dispersion, but to establish which physical inference is supported by the data and which remains limited by calibration, resolution, or selection.

Extending the framework to cosmic dawn

The same component-separation problem appears when the question moves from nearby host populations to the first galaxies and quasars. I contributed to a JWST/NIRCam study of AGNs at $7.15 \leq z \leq 7.75$ [5]. We decomposed rest-frame ultraviolet images into unresolved and extended components in a high-completeness sample. Assigning all unresolved light to AGNs gives a conservative upper bound, including compact sources that might instead contain stellar emission. Even with this assignment, AGNs supply at most about one third of the ionizing photons required at $z \simeq 7.5$.

This result extends the same logic from host measurements to a population-level constraint on the cosmic photon budget. Taken together, my projects form one research programme: develop component-aware measurements; use imaging, spectroscopy, variability, and forward models to test them; calibrate the black-hole and stellar-kinematic scales; and then use the resulting framework to study how nuclear growth is connected to host structure across cosmic time. This trajectory naturally leads to spatially

resolved near-infrared spectroscopy of the earliest quasars, where host galaxies, extended gas, and nuclear accretion can be studied within one common analysis framework.

References

[1] Sun, S. et al. 2026, *ApJ*, 1004, 13, Quiescent Host Galaxies of Extended Quasars.

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[4] Sun, S., Jiang, L., Bosman, S. E. I., & Cheng, Y., Stellar Velocity Dispersions of Quasar Host Galaxies with a Neural Network Trained on Forward-Modelled Spectra, in preparation.

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